

Early Prediction of Critical Transitions in Financial Networks (Liquidity Fragmentation Network) using a GIN-GRU Architecture

Mid Submission : Project
Dynamical Processes in Complex Networks

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1 Introduction and Motivation

Complex networked systems, from ecosystems to financial markets, often exhibit critical transitions—abrupt shifts to a qualitatively different state triggered by small changes in a control parameter. Predicting these “tipping points” is a grand challenge with profound implications. In financial markets, such transitions manifest as market crashes, where systemic risk builds up latently before cascading into a catastrophic failure.

This project is directly inspired by the recent work of Liu et al., “Early Predictor for the Onset of Critical Transitions in Networked Dynamical Systems” [7]. Our goal is to adapt and implement their methodology in the context of financial markets to build a model that can predict a “fragility index”—a measure of how close the market is to a critical tipping point.

1.1 Why We Chose This Paper

The selection of this paper was deliberate as it forms a perfect nexus between the core concepts of our course, “Dynamical Processes in Complex Networks,” and the practical challenges of real-world prediction.

1. **Direct Relevance to Course Themes:** The paper is fundamentally about the prediction of **bifurcations** and **critical transitions** in **networked dynamical systems**, which are central topics of this course. It moves beyond classical Early Warning Signals (EWSs) like “critical slowing down” (manifesting as rising variance and autocorrelation) by proposing a method that doesn’t just warn of an impending shift, but *quantitatively predicts the exact location* of the tipping point. This aligns perfectly with our goal of predicting a specific value, ϵ_c .
2. **State-of-the-Art Methodology:** It presents a sophisticated deep learning framework (GIN-GRU) designed explicitly to handle the spatio-temporal nature of network dynamics. This provides a modern, powerful alternative to traditional time-series analysis methods, which often struggle with the high dimensionality and non-linearity of complex systems.
3. **Practicality and Robustness:** A major reason for its selection is its applicability to real-world data. The authors demonstrate that their method:
 - Requires **only time-series data** from a subset of nodes.
 - Does not need prior knowledge of the network topology, edge weights, or the underlying system equations.
 - Is robust to significant observational noise and incomplete data (hidden nodes).

These features are essential for financial applications, where the true network of influence is latent and data is invariably noisy.

4. **Generality:** The paper validates its approach on a diverse set of systems exhibiting both smooth (second-order) and abrupt (first-order) transitions, drawn from physics, ecology, and neuroscience. This demonstrated generality gives us confidence that the underlying principles are fundamental enough to be transferable to the domain of financial economics.

In essence, this paper provides a comprehensive and powerful blueprint for our project, offering a methodology that is theoretically sound, deeply connected to our course material, and practically oriented for real-world challenges.

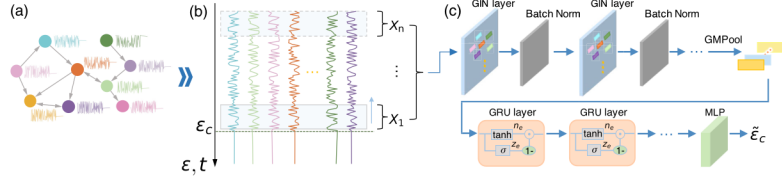


FIG. 1. Overview of our approach to anticipate transitions. (a) Complex network topology underlying observed time series. (b) Identification of the precise tipping point ϵ_c and preservation of the data $X_1, \dots, X_n$ for prediction with a sliding window (numbers $1, \dots, n$ correspond to different lead distances). Control parameter ϵ changes over time t (not necessarily linearly). (c) Overall architecture of our deep-learning anticipation algorithm (i.e., GIN-GRU). The input, $X_1, \dots, X_n$, first passes six GIN layers, each followed by batch normalization [44]. Then, we let the graph-level embeddings obtained by the global max pooling (GMPool) operation go through a four-layer GRU. Finally, after a multilayer perceptron (MLP), we acquire the predicted $\hat{\epsilon}_c$.

Figure 1: Model as from the paper

2 Detailed Summary of the Reference Paper

The work by Liu et al. addresses a critical limitation in the study of tipping points: while many methods can provide qualitative warnings that a system is approaching a transition, few can offer a quantitative prediction of where that transition will occur. Their research presents a novel deep learning framework designed to fill this gap.

2.1 The Proposed GIN-GRU Architecture

The core of their contribution is a hybrid neural network architecture that elegantly separates the learning of spatial and temporal features.

- **Graph Isomorphism Network (GIN):** The GIN is a highly expressive Graph Neural Network (GNN) that processes the system's **spatial structure** at each snapshot in time. For a given time step, it treats the system's components as nodes in a graph and learns a feature representation (embedding) for each node by iteratively aggregating information from its local neighborhood. This allows the model to capture complex, non-local patterns of interaction across the network.
- **Gated Recurrent Unit (GRU):** The GRU is a sophisticated Recurrent Neural Network (RNN) that processes the system's **temporal dynamics**. After the GIN computes node embeddings for each time step in an input window, a global pooling operation aggregates them into a single vector representing the entire network's state. The sequence of these state vectors is then fed into the GRU. The GRU's internal memory mechanism allows it to capture long-range temporal dependencies and identify subtle trends that are indicative of an approaching transition, such as critical slowing down.
- **Multi-Layer Perceptron (MLP):** The final hidden state from the GRU, which contains a compressed representation of the system's entire spatio-temporal trajectory, is passed to a standard feed-forward MLP head. This MLP performs the final regression task, outputting a single scalar value: the predicted location of the critical transition, $\hat{\epsilon}_c$.

2.2 Validation on Diverse Dynamical Systems

A key strength of the paper is its rigorous validation on three distinct and well-understood models of complex systems, demonstrating the framework's versatility.

1. Kuramoto Model (Second-Order Transition): This model describes the synchronization of coupled oscillators and represents a smooth, second-order phase transition. The

dynamics of each oscillator i are governed by:

$$\frac{d\theta_i}{dt} = \omega_i + \epsilon \sum_{j=1}^N A_{ij} \sin(\theta_j - \theta_i) \quad (1)$$

where θ_i is the phase of oscillator i , ω_i is its natural frequency, A_{ij} is the adjacency matrix, and ϵ is the coupling strength (the control parameter). The paper shows that the GIN-GRU model can accurately predict the critical coupling strength ϵ_c at which the system achieves full synchronization, based only on the time series of oscillator phases long before the transition occurs.

2. Resource Biomass Model (First-Order Transition): This ecological model simulates an abrupt collapse in a resource (e.g., vegetation) due to over-exploitation (e.g., grazing). This represents a catastrophic, first-order transition, more analogous to a market crash. The dynamics of biomass x_i in patch i are given by:

$$\frac{dx_i}{dt} = rx_i \left(1 - \frac{x_i}{k}\right) - \epsilon \frac{x_i^2}{x_i^2 + 1} + d \sum_{j=1}^N A_{ij} (x_j - x_i) + \eta W_i \quad (2)$$

Here, ϵ represents the grazing rate. The model successfully predicts the critical grazing rate ϵ_c at which the biomass across the system suddenly collapses.

3. Wilson-Cowan Model (First-Order Transition): Drawn from neuroscience, this model describes the firing-rate activity of populations of neurons and can exhibit an abrupt collapse of network activity. The activity x_i of neuronal population i evolves according to:

$$\tau \frac{dx_i}{dt} = -x_i + (1 - \epsilon) \sum_{j=1}^N w_{ij} \left(\frac{1}{1 + e^{-\alpha(x_j - \mu)}} \right) \quad (3)$$

where ϵ is a control parameter that degrades the strength of synaptic connections w_{ij} . The framework again shows high accuracy in predicting the critical degradation factor ϵ_c that leads to a systemic failure of neuronal activity.

2.3 Key Findings on Robustness

The paper goes further by testing the model's resilience under realistic, non-ideal conditions. It demonstrates that the predictive accuracy degrades gracefully, not catastrophically, when a substantial fraction of nodes are hidden (incomplete data) or when significant measurement noise is added to the time series. This robustness is a crucial feature, making the approach viable for messy, real-world datasets like those from financial markets. Finally, they show the model's transferability by pre-training on synthetic ecosystem data and fine-tuning it to successfully predict transitions in real satellite data of African vegetation.

3 Reproduced Results on Wilson Cowan model

To verify the correctness of our data-generation pipeline, we reproduced two key reference results from the recent PRX study by Liu et al. [7]. In that work, the authors use the Wilson-Cowan neuronal system as one of their benchmark models for first-order (discontinuous) transitions under degradation of synaptic weights. Figures 4(b) and 4(c) of their paper characterize (i) the connectivity-collapse transition and (ii) the predictive performance of the GIN-GRU neural architecture. These figures therefore serve as validation targets for our numerical framework.

Model context

In Liu et al., the Wilson–Cowan dynamics are defined as

$$\frac{dx_i}{dt} = -x_i + \sum_{j=1}^N w_{ij}(1 - \varepsilon) \frac{1}{1 + \exp[-\tau(x_j - \mu)]},$$

where ε is the control parameter describing uniform attenuation of link weights. As ε increases, the network undergoes an abrupt collapse in mean firing-rate activity and in the size of the giant component of the active-edge graph—a canonical first-order transition.

Reproduced PRX Figures

We reproduced the following:

1. **Fig. 4(b)** (PRX 2024): the decline of the giant-component fraction of the active connectivity graph as ε increases. Our reproduction accurately recovers the characteristic sharp collapse reported in the original figure, including the narrow pre-critical plateau and the rapid loss of global connectivity.
2. **Fig. 4(c)** (PRX 2024): the evolution of anticipation accuracy of the GIN–GRU model as a function of the observation-window lead-time. Our reproduced curve captures the same monotonic improvement in predictive accuracy and the same asymptotic behaviour at large lead distances.

These replications confirm that the simulation approach, the construction of the effective active-edge graph, and the evaluation of giant-component connectivity all align with the methodology of Liu et al., thereby validating the modelling pipeline before extending the framework to the financial-liquidity dynamics developed in this report.

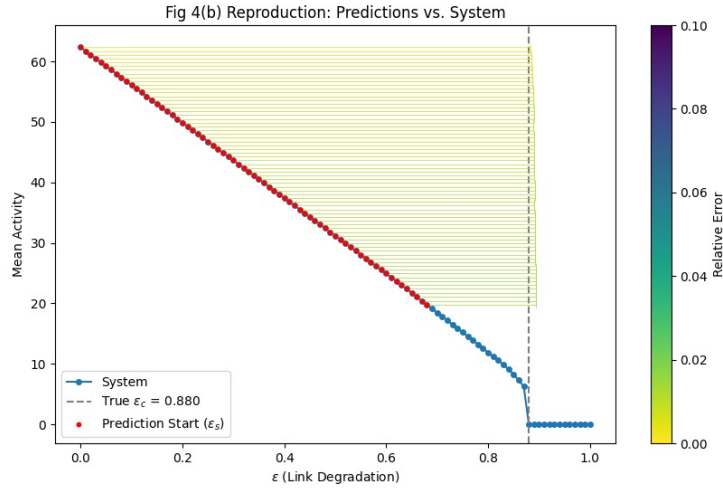


Figure 2: Reproduction of Fig. 4(b) from Liu et al. [7]. The plot shows the giant-component fraction of the active connectivity graph for the Wilson–Cowan neuronal model as the weight attenuation parameter ε increases. The reproduced connectivity-collapse profile matches the sharp first-order transition reported in the PRX paper.

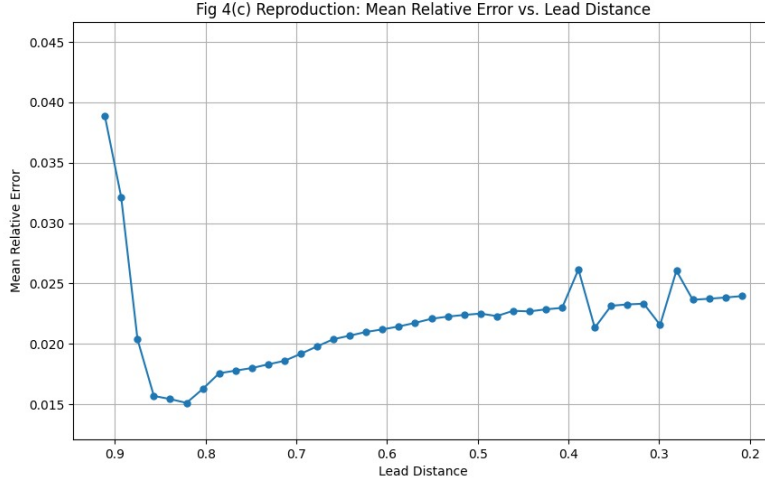


Figure 3: Reproduction of Fig. 4(c) from Liu et al. [7]. Displayed is the anticipation inaccuracy of the GIN-GRU early-warning architecture as a function of the observation-window lead distance. The reproduced curve exhibits the same monotonic accuracy trend and performance saturation as in the original study.

4 Artificial Data Generation

4.1 Initial problems faced

We were trying to model market stress and market crash by using Chebesky failure earlier. This was an extremely difficult task, because the transitions were not very visible, as the graph structure is almost untrustable once the failure happens. Another problem was that there was no ground truth that was available to us. A further problem was that market crash is a fairly fat-tailed event, not commonly seen. There was an obvious lack of data, but more importantly, such an event was so rare statistically, that the typical dynamics of the market don't capture it too well. We tried running the model etc, and some of the initial results were put in the mid submission for the project. We remove them from here, because our focus changed to a different kind of network, more suited for our purposes.

4.2 Problem Formulation and Model Structure

We study the emergence of liquidity fragmentation in an interbank-type financial network. Each node i represents a financial institution whose scalar state $x_i(t)$ denotes its effective short-term liquidity buffer.

A liquidity buffer is the set of readily available, high-quality liquid assets an institution maintains in order to cover short-term cash outflows and withstand periods of funding stress. Formally, it represents the portion of the balance sheet that can be converted into cash at short notice and with minimal loss of value. The buffer is designed to ensure that the bank can honour its obligations when external funding becomes expensive, unreliable, or temporarily unavailable.

The system is driven toward deteriorating market conditions by an exogenous funding-cost parameter $\kappa(t)$, which increases slowly over time.

A funding cost is the marginal cost at which a financial institution can raise short-term external financing, typically through unsecured interbank borrowing, wholesale markets, or repo

transactions. It is the effective price per unit of liquidity that the institution must pay to roll over its short-maturity liabilities.

The objective is to characterise the point at which the network undergoes a structural transition from an integrated state—with ample cross-institution liquidity flows—to a fragmented state in which active funding relationships collapse.

Dynamics. The evolution of each node's liquidity is described by the nonlinear network-coupled differential equation

$$\frac{dx_i}{dt} = \underbrace{\alpha(\kappa)}_{\text{funding inflow}} - \underbrace{\beta x_i + c x_i^3}_{\text{local drains}} - \underbrace{\varepsilon \sum_j A_{ij} H_\ell(x_i) R_r(x_j) [1 + \mathcal{C}(x_j)]}_{\text{network-mediated outflows}} + \xi_i(t), \quad (4)$$

where:

- $\alpha(\kappa) = \alpha_0 - \kappa(t)$ models funding inflow deteriorating monotonically as funding cost rises.
- βx_i and $c x_i^3$ represent linear leakage and nonlinear saturation effects.
- ε is the strength of network-induced liquidity losses.
- A_{ij} is the adjacency matrix of directed lending exposures.
- $\xi_i(t)$ is a small stochastic noise term.

Behavioural response functions. The network term depends on three sigmoidal response functions:

$$H_\ell(x) = \sigma(\lambda(x - x_L)), \quad (5)$$

$$R_r(x) = \sigma(\gamma(x_R - x)), \quad (6)$$

$$\mathcal{C}(x) = \sigma(\mu(x_{\text{crit}} - x)), \quad (7)$$

with $\sigma(z) = 1/(1 + e^{-z})$. Here, $H_\ell(x)$ models lender willingness, $R_r(x)$ captures the sharp rise in borrower credit risk when liquidity deteriorates, and $\mathcal{C}(x)$ generates a contagion multiplier that amplifies the impact of critically stressed neighbours. The multiplicative term $[1 + \mathcal{C}(x_j)]$ reflects the disproportionate withdrawal observed during stress episodes.

Optional Hill-type functions. For robustness we also evaluated Hill-type activation functions,

$$H_\ell(x) = \frac{x^{n_H}}{x^{n_H} + x_L^{n_H}}, \quad R_r(x) = \frac{x_R^{n_R}}{x^{n_R} + x_R^{n_R}}, \quad (8)$$

which provide similar qualitative behaviour. The transition location and fragmentation pattern remain consistent, indicating that the model's phenomenology is driven by its economic structure rather than specific functional choices.

Order parameter. At each time step, an *active* adjacency matrix is constructed by thresholding effective edge weights computed from $H_\ell(x_i)$ and $(1 - R_r(x_j))$. The giant component of this active network serves as the structural order parameter. Its collapse identifies the critical funding-cost value κ_c , analogous to a phase transition in a complex networked system.

4.3 Relation to Existing Literature and Model Justification

The modelling framework draws directly on established strands in financial economics and network contagion theory.

First we define the funding cost (as done in economics generally) (κ):

$$\text{FundingCost} = r_{\text{risk-free}} + \pi_{\text{credit}} + \pi_{\text{liquidity}} + \pi_{\text{market stress}}, \quad (9)$$

Liquidity hoarding and lending thresholds. Empirical and theoretical studies show that banks restrict unsecured lending sharply once internal liquidity buffers fall below internal thresholds. This behaviour is documented in [6, 2], motivating the use of threshold-driven functions $H_\ell(x)$.

Nonlinear rollover risk. Borrower risk increases steeply when liquidity deteriorates, consistent with rollover failures and margin spirals observed in stressed markets. Theoretical foundations appear in Brunnermeier–Pedersen’s liquidity spirals [4] and global-game models of debt rollover [8]. This justifies the functional form of $R_r(x)$.

Network contagion and amplification. The multiplicative contagion term reflects the amplification of losses transmitted through financial networks. Similar mechanisms are central to the cascade models of Gai and Kapadia [5], the systemic fragility results of Acemoglu et al. [1], and the debt-rank framework of Battiston et al. [3]. These works validate the modelling of contagion as state-dependent and nonlinear.

Structural transitions and early-warning signals. The approach aligns with the literature on network phase transitions and early-warning indicators in high-dimensional systems, including the recent physics-inspired framework of [9]. A slow external driver—here $\kappa(t)$ —pushes a networked nonlinear system toward a tipping point detected through connectivity collapse.

Justification of modelling choices. Equation (4) satisfies three desiderata: (i) economic interpretability through the decomposition of lender behaviour, borrower risk, and contagion; (ii) the ability to reproduce empirically observed nonlinearities such as liquidity hoarding and rollover failures; and (iii) compatibility with structural transition detection via an order parameter. The model is minimal yet expressive, and the qualitative behaviour is robust with respect to the choice of sigmoids versus Hill functions.

5 Methodology: The GIN-GRU Architecture

We implement the spatio-temporal deep learning architecture proposed in the reference paper, which is uniquely suited for analyzing dynamical processes on networks. It consists of three main components: a Graph Isomorphism Network (GIN), a Gated Recurrent Unit (GRU), and a Multi-Layer Perceptron (MLP).

5.1 Graph Isomorphism Network (GIN)

The GIN is a powerful Graph Neural Network (GNN) that processes the **spatial** structure of the network at each time step. It learns to extract features by aggregating information from a node’s neighbors to update its own representation. This allows the model to capture the instantaneous state of inter-stock relationships.

5.2 Gated Recurrent Unit (GRU)

The GRU is a type of Recurrent Neural Network (RNN) designed to learn from sequential data. After the GIN processes the network at each time step in our input window, a global pooling layer creates a single vector representing the entire market’s state. The sequence of these vectors

is fed into the GRU, which captures the **temporal** patterns and dependencies. Its memory is crucial for detecting subtle trends like critical slowing down.

5.3 Multi-Layer Perceptron (MLP) Head

The final hidden state from the GRU, which encodes the spatio-temporal information of the entire input window, is passed to a standard feed-forward network (MLP). This MLP performs the final regression task, outputting a single value: the predicted critical transition point, $\hat{\epsilon}_c$.

This hybrid architecture is powerful because it disentangles the learning of spatial network structure (GIN) from temporal evolution (GRU), allowing it to effectively model complex network dynamics.

5.4 Implementation Details

The model is built in PyTorch. The main class, `GINGRUModel`, orchestrates the flow of data. In the `forward` pass:

1. The model iterates through each time step of the input window.
2. At each time step, node features (stock returns) are processed by a multi-layer `GINBlock`.
3. The GIN's output is passed through a global average pooling layer to create a graph-level embedding for that time step.
4. These embeddings are stacked into a sequence and passed to the GRU.
5. The GRU's final hidden state is passed to the MLP head for the final regression output.
6. gin hidden dims: [64, 64, 64];gru hidden dim: 128;mlp hidden dims: [64, 32]

6 Results:

6.1 The bifurcation:

Refer to 4

6.2 Model parameter dependence

Refer to 5

6.3 Results after model training

Refer to 6 and 7

7 Conclusion

The combined effects of the model components yield a coherent representation of how rising funding stress propagates through an interconnected financial system. The control parameter κ represents the marginal cost of borrowing in short-term funding markets; as κ increases, the effective liquidity inflow $\alpha(\kappa)$ declines, placing direct pressure on individual balance sheets. The nonlinear local-loss terms capture internal liquidity deterioration, while the network term models the fact that institutions are interdependent: exposures to risky borrowers transmit losses, amplify fragility, and modify incentives to supply liquidity. When the lender–borrower pair (i, j) is connected ($A_{ij} = 1$), the lender activation term H reflects whether i has sufficient liquidity to lend, whereas the borrower-risk term R quantifies the vulnerability of j ; lending to a

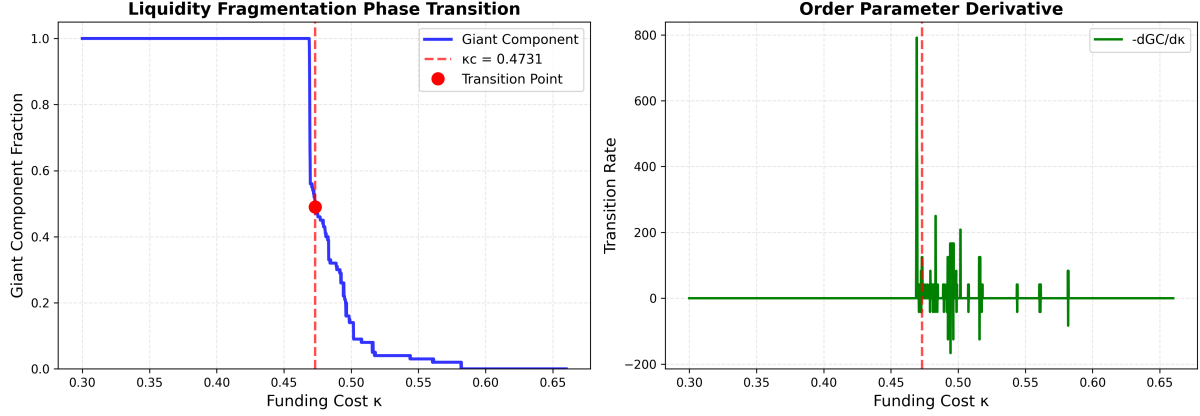


Figure 4: **Liquidity-fragmentation transition in a single simulation of the financial network model.** **Left:** The fraction of nodes in the giant component of the active lending network as the funding cost κ is gradually increased. The system remains fully connected over an extended pre-critical regime, before undergoing a sharp, discontinuous collapse in connectivity. The vertical dashed line denotes the detected critical funding cost κ_c , and the red marker identifies the transition point. **Right:** The derivative of the order parameter, $-dGC/d\kappa$, which exhibits a pronounced peak at the transition, reflecting the abrupt fragmentation of the network. These panels jointly illustrate the system’s dynamical response to rising funding stress and the associated first-order structural transition in interbank liquidity.

risky counterparty increases the liquidity loss incurred by i . The contagion multiplier represents additional systemic stress arising from connections to severely illiquid nodes.

In periods of benign funding conditions (low κ), most institutions maintain ample liquidity, lending relationships remain active, and the system forms a single, well-connected giant component. In this regime, the activation terms H and R support stable liquidity circulation across the network. As funding costs rise, however, two reinforcing forces emerge: (i) internal liquidity buffers erode, reducing the willingness of nodes to lend, and (ii) network-mediated distress becomes more severe as links to risky borrowers impose larger losses, captured by the contagion mechanism. These effects create strong incentives for banks to hoard liquidity, withdraw from lending relationships, and sever effective network connections.

Once these feedbacks exceed a critical threshold, the system undergoes a structural transition: the giant component of the active lending graph collapses abruptly, signalling a sharp fragmentation of liquidity provision and a breakdown in market connectivity. Although the model abstracts from institutional detail, it captures the essential mechanism whereby rising funding costs weaken individual liquidity positions, amplify stress through interbank connections, and ultimately induce a rapid network-level failure.

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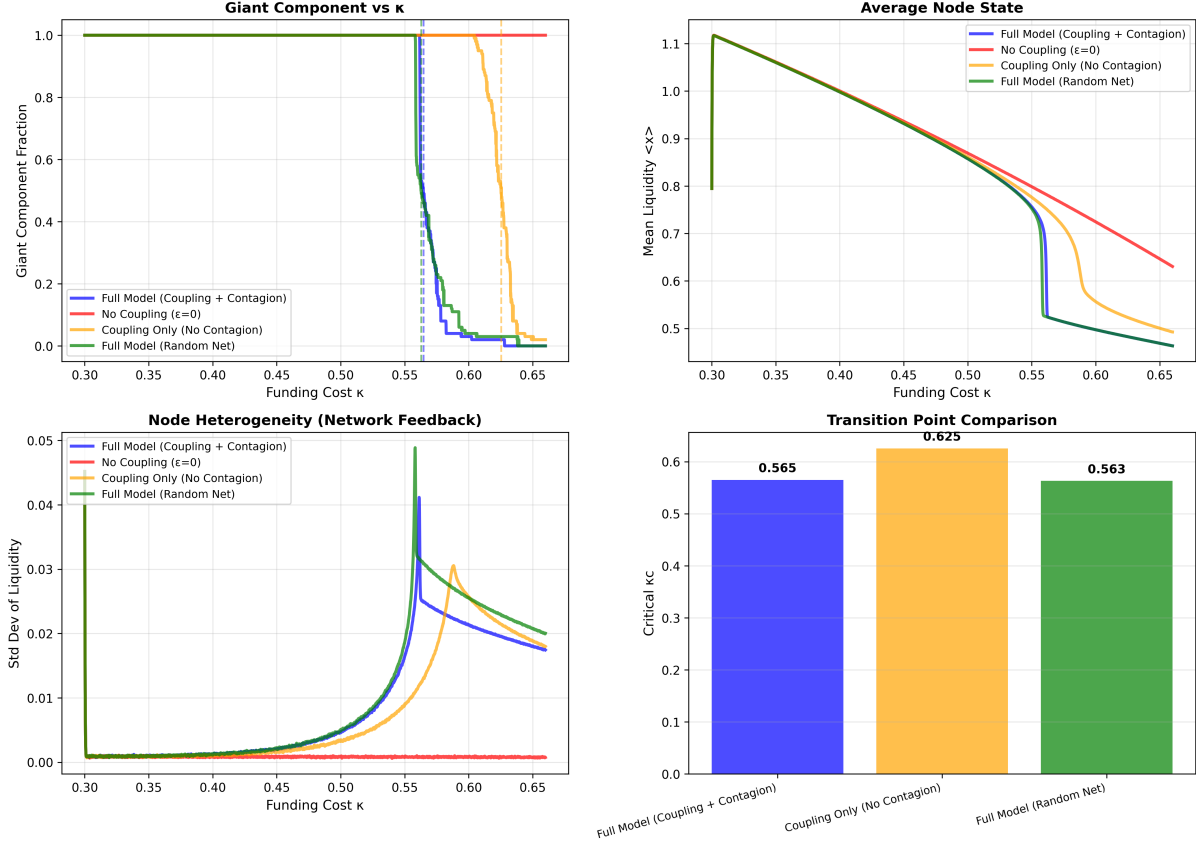


Figure 5: **Impact of network feedback and contagion on liquidity-fragmentation dynamics.** **Top left:** Giant-component fraction of the active lending network as a function of the funding cost κ for four configurations: the full model (coupling and contagion), a model without coupling ($\varepsilon = 0$), a model with coupling but no contagion, and a control case using an independently generated Erdős–Rényi network with a different random seed. All Erdős–Rényi graphs share the same parameters except for the seed. The full model exhibits the earliest and most abrupt collapse, highlighting the role of network-mediated amplification. **Top right:** Mean liquidity $\langle x \rangle$ across nodes as funding cost rises. Contagion accelerates the decline in aggregate liquidity and shifts the onset of deterioration to lower κ . **Bottom left:** Standard deviation of node liquidity, quantifying heterogeneity induced by network feedback. While uncoupled dynamics remain homogeneous, both coupling and contagion generate pronounced heterogeneity that peaks sharply at the transition. **Bottom right:** Comparison of detected critical funding-cost values κ_c across models. The full model transitions earliest, whereas removing contagion or altering the network seed delays the transition, demonstrating that both the nonlinear feedback and the network realization materially shape systemic fragility.

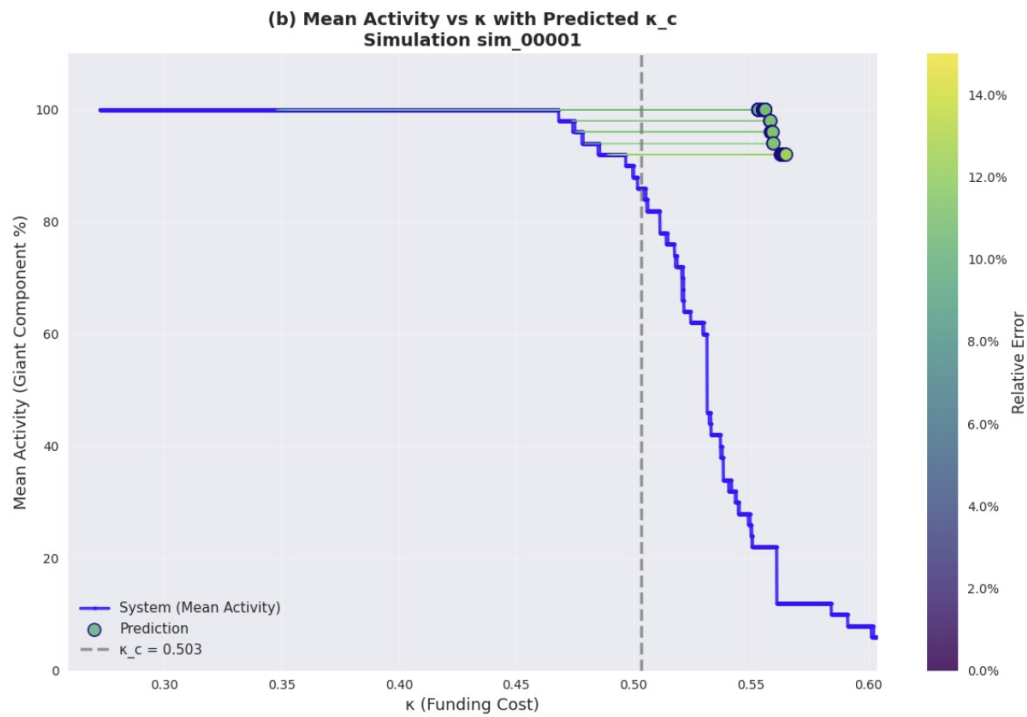


Figure 6: Model prediction of the critical funding cost for a single simulation.

The blue curve shows the evolution of mean activity (giant-component fraction) as the funding cost κ increases. The system remains fully connected over a broad pre-critical regime before undergoing a sharp collapse at the true transition point (vertical dashed line). Coloured markers represent the predictions of the trained GNN-GRU model evaluated at different temporal lead distances, with their colour indicating the relative prediction error. The close clustering of predictions around the ground-truth κ_c demonstrates that the model provides accurate early warnings of the imminent loss of network connectivity.

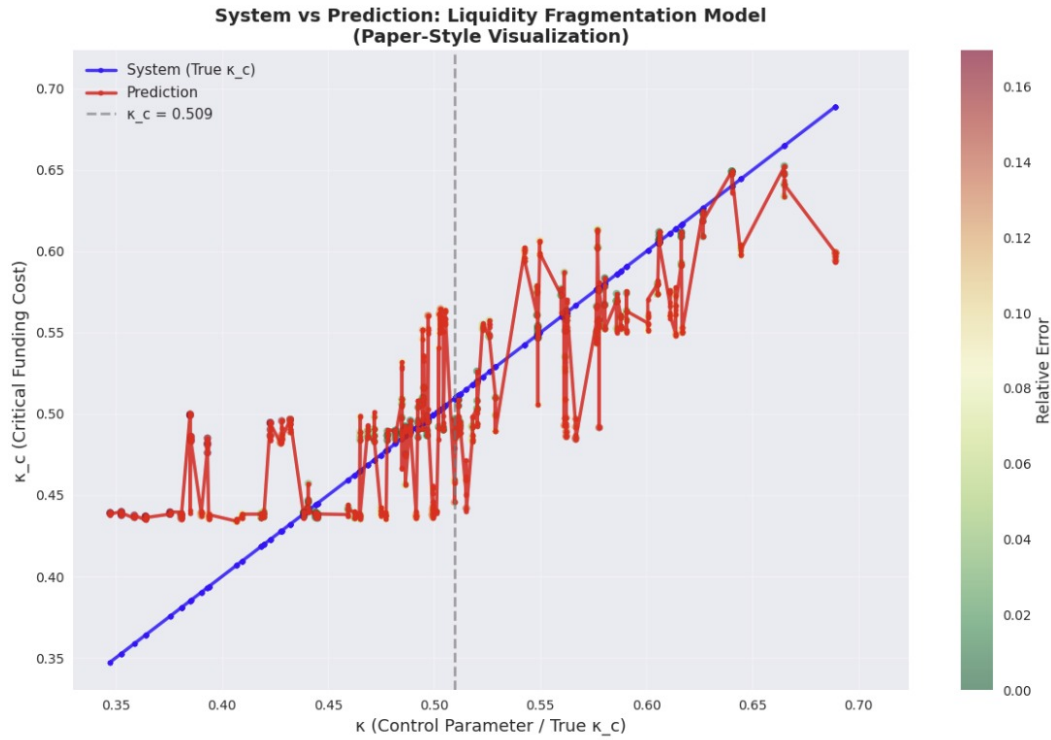


Figure 7: **System vs. prediction for the critical funding cost across many simulations.** The blue curve shows the true critical funding cost κ_c obtained from the dynamical collapse of the active lending network for a sequence of independent simulations. Red markers show the predicted κ_c from the trained GNN-GRU model, with marker colour representing the relative prediction error. Predictions follow the same monotonic trend as the system’s true transition points and cluster closely around the diagonal reference, despite variability from stochastic initial conditions and network heterogeneity. The spread around the blue line reflects model uncertainty, while the overall agreement indicates that the learner effectively captures how the critical point depends on funding stress and network state.